PROGRAM IMPLEMENTATION PLAN · CONFIDENTIAL DRAFT

AutoOBE Implementation Plan

CO–PO Mapping & Attainment Platform for NBA Accreditation — built to run both the legacy 12-PO (GAPC v3) and current 11-PO (GAPC v4.0) frameworks concurrently, from day one.

Prepared by	Program Management Office (20+ yrs enterprise SaaS & platform launches, ex-Google PM practice)
Prepared for	Founding engineering, curriculum, and compliance workstream leads
Based on	AutoOBE v3 Engineering & GTM Plan (verified against the official NBA SAR, GAPC v4.0, Jan 2025)
Status	Draft for kickoff review — v1.0
Distribution	Internal — program team, sponsor, steering committee

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Section 1

Program overview & objectives

Mission. Ship a platform that generates defensible CO–PO mappings and NBA attainment reports automatically, with every faculty touchpoint reduced to a one-click approval — and do it in a way that never forces a customer college to choose between their historical 12-PO records and their current 11-PO obligations.

Why this program exists now. NBA's GAPC v4.0 transition (effective Jan 2025, Tier-II transition window closed June 2025) means every affiliated college sitting on three-plus years of 12-PO matrices in Excel needs a path forward that doesn't discard that history. That migration problem is the wedge into a market where two competitors already run a similar playbook (see the source plan, §4) — execution speed and framework correctness are the entire competitive advantage. This is a program-management problem as much as an engineering one: sequencing, staffing, and go/no-go discipline decide whether we ship the 12→11 bridge before or after a competitor does.

Objectives and key results — first two quarters

Objective 1 (Q1) — Prove the deterministic core is correct and dual-framework-safe before any AI is layered on top.

- KR1: 100% of PO/CO schema objects are keyed by `(framework_version, po_code)` — zero hardcoded PO counts anywhere in code, verified by a static-analysis check in CI.
- KR2: Attainment engine reproduces both a hand-computed 12-PO example and an 11-PO example from real AKTU data, bit-for-bit, in the regression suite.
- KR3: 100 free-tier faculty signups on the zero-AI Phase-1 tool (upload QP + marks → CO attainment report).

Objective 2 (Q2) — Make the mapping engine trustworthy enough that a department head bulk-approves it in one sitting.

- KR1: ≥85% of generated CO–PO mapping rows land in the auto-accept confidence band (≥0.85) on the ~500-pair gold set.
- KR2: 3 pilot departments complete a full course-matrix review in under 2 hours each (target: 90 seconds per course).
- KR3: 12→11 migration tool converts a real pilot college's legacy 12-PO matrix set with zero data loss, verified against the college's own historical Excel sheets.

OKRs for Q3 onward (ingestion depth, indirect assessment, SAR assembly) are set at the end of Q2 based on what Q1–Q2 actually teaches us — consistent with running this as a staged program, not a fixed waterfall.

Section 2

Scope

IN SCOPE	DEFINITION OF DONE
SAR Criteria 1, 3, 8 (full)	All sub-items generate a populated, exportable SAR table matching the official template structure exactly (Tables 1.4.2.1, 1.5.1, 3.8.1–3.8.3, etc.).
Outcome-mapping portions of Criterion 2	Capstone/complex-problem/SDG evidence linkage populated where it depends on CO–PO data already captured.
Dual framework support (12-PO / 11-PO)	A program can run historical cohorts under GAPC v3 and current cohorts under GAPC v4.0 simultaneously, with a documented crosswalk and a working migration tool. This is the subject of Section 3.
AKTU curriculum corpus + at least one additional scheme	Structured, versioned, K-level-array-correct course/CO records available as a subscribed library, not re-typed per college.

OUT OF SCOPE (THIS PROGRAM)	REASON
SAR Criteria 5, 6, 7, 9 (faculty records, facilities, governance)	440 marks of ERP-territory work already owned by incumbents; adds surface area, not differentiation.
Full LMS, fee management, timetabling, admissions	Explicitly not the product. Integrate with the college's ERP; don't become one.
NAAC / NIRF modules	Different accreditation body, different data model, different buyer — a future program, not this one.

Section 3

Dual program-outcome framework — the core requirement

This is the one requirement in the entire program that, if we get it wrong, cannot be patched later without a data migration on every customer's live attainment history. It gets its own section, its own full text, and its own explicit sign-off gate at the end of Phase 0.

3.1 Why both frameworks must live in the system at once

NBA's own transition guidance is that a college's *already-completed* batches do not need to be retroactively re-mapped — but any attainment figure a college presents going forward must be in the current 11-PO structure. In practice this means, for the next several years, a single institution will have:

- Final-year and graduated cohorts whose entire CO–PO matrix, and years of accumulated attainment history, was authored under the **old 12-PO (GAPC v3)** framework.
- Current and incoming cohorts whose curriculum, mapping, and attainment must be authored under the **new 11-PO (GAPC v4.0)** framework.
- A continuing need to show trend lines (year-over-year PO attainment) across the transition boundary for Criterion 8's two-year closed-loop improvement evidence — which is impossible without a documented crosswalk between the two frameworks.

> **Program decision:** both frameworks are first-class, permanently — not a temporary shim to be deleted after a migration window. Treat this the way a platform team treats API versioning: old and new coexist, each fully supported, for as long as any customer has live data on the old version.

3.2 GAPC v3 — the twelve Program Outcomes (verbatim, legacy framework)

PO	TITLE	STATEMENT (VERBATIM)
PO1	Engineering Knowledge	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO3	Design/Development of Solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.
PO4	Conduct Investigations of Complex Problems	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO5	Modern Tool Usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO	TITLE	STATEMENT (VERBATIM)
PO6	The Engineer and Society	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	Environment and Sustainability	Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for, sustainable development.
PO8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
PO9	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large — comprehend and write effective reports and design documentation, make effective presentations, give and receive clear instructions.
PO11	Project Management and Finance	Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects in multidisciplinary environments.
PO12	Life-Long Learning	Recognize the need for, and have the preparation and ability to engage in, independent and lifelong learning in the broadest context of technological change.

Verified against a published NBA-accredited institution's official PO document (cross-checked for verbatim consistency with the standard NBA/Washington Accord Graduate Attributes text used nationally pre-2025). Re-confirm against the specific archived NBA circular before this text is embedded byte-for-byte in production, per the same rule applied to the 11-PO text.

3.3 GAPC v4.0 — the eleven Program Outcomes (verbatim, current framework)

PO	TITLE	WK LINKAGE	STATEMENT (VERBATIM)
PO1	Engineering Knowledge	WK1–WK4	Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization… to develop the solution of complex engineering problems.
PO2	Problem Analysis	WK1–WK4	Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
PO3	Design/Development of Solutions	WK5	Design creative solutions for complex engineering problems… with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment.
PO4	Conduct Investigations of Complex Problems	WK8	Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data.

PO	TITLE	WK LINKAGE	STATEMENT (VERBATIM)
PO5	Engineering Tool Usage	WK2, WK6	Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations.
PO6	The Engineer and The World	WK1, WK5, WK7	Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability...
PO7	Ethics	WK9	Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.
PO8	Individual and Collaborative Team Work	—	Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	Communication	—	Communicate effectively and inclusively within the engineering community and society at large...
PO10	Project Management and Finance	—	Apply knowledge and understanding of engineering management principles and economic decision-making...
PO11	Life-Long Learning	WK8	Recognize the need for, and have preparation and ability for independent and life-long learning, adaptability, and critical thinking.

Transcribed directly from Annexure II of the official NBA SAR, UG Engineering Tier-II, GAPC v4.0, January 2025 (nbaind.org). Byte-exact.

3.4 Crosswalk — what the schema and the migration tool must encode

GAPC V3 (12-PO)	GAPC V4.0 (11-PO)	TRANSITION RULE
PO1–PO5	PO1–PO5	Direct 1:1 carry-forward (PO5 renamed “Modern Tool Usage” → “Engineering Tool Usage”, same slot).
PO6 (The Engineer and Society) + PO7 (Environment and Sustainability)	PO6 (The Engineer and The World)	**Merge.** Historical attainment for old PO6 and old PO7 must both roll up into new PO6 for any cross-framework trend view — store both source values, compute a documented blended figure, never silently discard one.
PO8–PO12	PO7–PO11	**Shift down one slot.** Every downstream reference (mapping rows, attainment rows, survey PO columns) keyed by ordinal position breaks here — this is exactly why the schema key is `po_code` within a `framework_version`, never a bare integer index.

> **Engineering acceptance criterion for this section (Phase 0 exit gate):** a single program record can hold CourseOfferings tagged `framework_version=GAPC_V3` and others tagged `GAPC_V4` in the same academic year, each computing attainment independently and correctly against its own 12 or 11 columns, with the crosswalk table available as a reporting view — not a data transformation that overwrites history.

Section 4

Workstreams & RACI

R = Responsible, A = Accountable, C = Consulted, I = Informed. One accountable owner per row, always — the single most common cause of a stalled program is a deliverable with two accountable owners or none.

DELIVERABLE	PROGRAM LEAD	BACKEND/PLATFORM	ML/DATA	FRONTEND	CURRICULUM LEAD	COMPLIANCE/DOMAIN LEAD
Dual-framework schema & crosswalk (§3)	A	R	C	I	C	C
AKTU syllabus corpus (structured, K-level arrays)	A	I	R	I	R	I
SQL attainment engine, both target modes	I	R/A	C	I	I	C
Mapping engine, layers 1–8	I	C	R/A	I	C	C
12→11 migration tool	A	R	C	C	C	R
Spreadsheet-grid UI, Excel paste	I	C	I	R/A	I	I
DPDP DPA template & minor-consent flow	C	C	I	C	I	R/A
SAR export templates (Tables 1.4.2.1 … 3.8.3)	I	R	I	C	C	A
Free-tier launch & faculty acquisition	R/A	C	I	C	I	I

Section 5

Phased rollout & launch-readiness gates

Modeled on a staged-rollout discipline — each phase ends with a Launch Readiness Review (LRR): a short, written go/no-go decision against explicit exit criteria, not a calendar date. A phase does not start consuming the next phase's budget until its own gate is passed.

PHASE	WEEKS	FOCUS	EXIT CRITERIA (LRR GATE)
0 — Foundations	1–4	Corpus, gold set, dual-framework schema, verbatim PO/WK encoding	GO/NO-GO Both frameworks demonstrably coexist per §3.4's acceptance criterion; ~500-pair gold set exists; AKTU corpus parsing pipeline runs end-to-end on a sample branch.
1 — Deterministic core	1–6	Attainment engine (zero AI), Excel/CSV ingest, free single-faculty tool	100 faculty signups; CO attainment + Course Articulation Matrix export matches a hand-computed reference case for both frameworks.
2 — Mapping engine (private beta)	5–12	Layers 1–8, WK derivation, constraint solver, confidence gating, 12→11 migration tool	≥85% auto-accept on gold set; 3 pilot departments complete a full matrix review in a single sitting; migration tool round-trips a real college's legacy data with zero loss.
3 — Ingestion depth	11–17	DOCX/PDF QP parser + quality audit, AKTU result ingest, 2 ERP connectors, rubric builder	QP parsing validated against ≥20 real internal papers across ≥3 pilot colleges; rubric evidence stored and linked to PO8/9/10.
4 — Indirect + CQI	16–21	Survey types, auto-dispatch, indirect attainment, CQI generation	Survey response rate ≥60% in at least one pilot cycle; CQI records auto-generated and faculty-editable.
5 — GA: SAR assembly	20–28	Criterion 3 first, then 1, then 8, then outcome parts of 2; audit-trail export	One pilot college produces a complete Criteria 1/3/8 SAR section from the platform with only a review pass, no manual rebuild; DPDP DPA signed with all pilot colleges.
6 — Scale	28–34	Multi-program, multi-campus, year-on-year comparison across the framework boundary, mock-audit checklist	Cross-framework trend view (§3.4) live for at least one college with both legacy and current cohorts.

Section 6

Milestone schedule

WEEK	MILESTONE	OWNER	DELIVERABLE EVIDENCE
2	Framework tables locked	Backend/Platform	Both PO tables (§3.2, §3.3) and WK1–WK9 loaded as versioned reference data; crosswalk table reviewed by Compliance/Domain Lead.
4	Phase 0 LRR	Program Lead	Written go/no-go memo against §5 exit criteria.
6	Free-tier public launch	Frontend + Program Lead	Upload-QP-and-marks → download-report flow live, zero AI.
9	First pilot department onboarded to mapping engine	ML/Data	Full course matrix generated with evidence bundles.
12	Phase 2 LRR — migration tool sign-off	Compliance/Domain Lead	Legacy 12-PO data for ≥1 real college migrated and verified against source Excel.
17	Phase 3 LRR — ingestion depth	Backend/Platform	≥3 colleges ingesting real question papers and marks.
21	Phase 4 LRR — CQI live	ML/Data	Auto-generated CQI record accepted and edited by a real faculty user.
28	GA — first complete SAR section shipped	Program Lead	Pilot college's Criterion 3 export accepted for internal use without manual rebuild.
34	Program closeout & retro	Program Lead	Blameless retrospective; OKRs for next program phase set.

Section 7

RAID log

Risks

RISK	SEVERITY	OWNER	MITIGATION
Dual-framework crosswalk gets built as an afterthought, not a schema primitive	HIGH	Backend/Platform	Phase 0 LRR explicitly gates on the §3.4 acceptance criterion before any mapping-engine work starts.
Peer team rejects AI-generated mappings as not faculty-owned	HIGH	ML/Data	Explicit per-row approval logged with name/timestamp; report language “faculty-approved, system-assisted,” never “AI-generated.”
Numbers don't reproduce on re-run across a framework version bump	HIGH	Backend/Platform	Version-pin model + engine + framework per academic year; freeze snapshots; regression suite over the gold set on every deploy.
Cross-tenant data leak	HIGH	Backend/Platform	Row-Level Security at the database layer; pen-test before first multi-college deployment.
DPDP Rules still maturing; Data Protection Board not yet operational	MEDIUM	Compliance/Domain Lead	Treat as an ongoing compliance-watch process; build to a Nov 2026 readiness target, track MeitY notifications monthly.
AKTU syllabi still pure L-T-P; NCrF conversion not done upstream	MEDIUM	Curriculum Lead	Build the L-T-P→NCrF converter as core Phase 1 scope, not deferred.
Competitor (Edhitch) already live with a similar GAPC v4.0-native positioning	MEDIUM	Program Lead	Differentiate on evidence-grounded weight traceability and the dual-framework migration tool — neither currently offered by any competitor checked.
Faculty won't upload data even with a free tier	MEDIUM	Frontend	Excel paste, remembered layouts, ERP sync. The free tool must be measurably faster than Excel or the thesis fails.

Assumptions

- Pilot colleges are willing to share real (anonymizable) legacy 12-PO matrices for migration-tool validation.
- At least one pilot college has cohorts spanning both frameworks concurrently, to validate §3.4 against real data rather than synthetic data alone.
- NBA does not revise the SAR format again mid-program (framework versioning in the schema is the mitigation if it does).

Issues (open at program kickoff)

- The “80:20 SEE:CIE worked example” cited in earlier planning material could not be located in the actual 62-page SAR document on re-verification — default blend weights must be presented as a configurable vendor default, not a quoted NBA example, until a source is found.
- Engineering-specific AKTU college count is ambiguous across sources (199 vs. 400+) — needs direct verification before it appears in any customer-facing sizing claim.

Dependencies

- Domain/Compliance Lead hire (NBA-visit experience) is a hard dependency for the Phase 0 LRR sign-off on §3 — do not let this hire slip past week 2.
- Pilot college legal sign-off on the DPDP DPA template is a hard dependency for the Phase 5 GA gate.

Section 8

Success metrics

METRIC	TARGET	WHY IT'S THE RIGHT METRIC
North Star: minutes from upload to accepted CO attainment report	< 10 min	Directly measures whether the product is actually faster than Excel — the entire GTM thesis depends on this being true, not assumed.
Mapping engine auto-accept rate	≥ 85%	The stated design target (§5, Phase 2 gate); below this, the "90-second review" promise breaks.
Legacy-to-current framework migration accuracy	100%	Zero tolerance — this is customer historical data, not a generated suggestion.
Weekly active faculty (free tier)	100 by wk 6	Validates the acquisition wedge before the paid mapping engine is built.
SDG / rubric evidence completeness per project course	≥ 90%	Rubrics are the only accepted evidence path for PO8/9/10 — incomplete evidence here breaks the SAR export.

Section 9

Governance & communication cadence

FORUM	CADENCE	ATTENDEES	PURPOSE
Engineering standup	Daily, 15 min	Eng workstreams	Blockers only. No status theater.
Cross-functional sync	Weekly	All workstream leads	Cross-dependency check, especially anything touching §3's shared schema.
Steering committee	Bi-weekly	Program Lead, Sponsor, Domain/Compliance Lead	Risk review, LRR decisions, budget checkpoints.
OKR review	Quarterly	Full program team	Grade the prior quarter's KRs honestly; set the next quarter's.
Blameless retrospective	End of each phase	Full program team	What broke, why, what changes — written down, not just discussed.

Section 10

Budget & resourcing

LINE ITEM	ESTIMATE	NOTE
Team: 2 backend, 1 frontend, 1 ML/data, 1 domain lead × 8 months	₹70–90L	2026 India salary bands; upper end if the ML/data hire needs genuine LLM/embeddings experience.
LLM inference (mapping engine)	₹1,000–1,700 / program / yr	~1,500 batched calls/program/yr at current Sonnet-tier pricing; negligible but must be budgeted correctly, not hand-waved.
Infrastructure (pilot scale)	< ₹1L / month	Postgres + Redis + object storage, single-region.
Domain lead advisory / consulting rate	₹1.5–3L / month if full-time	Not covered by standard tech-salary benchmarks; treat separately in the offer.

Section 11

Launch-readiness checklist (Phase 5 GA gate)

- [ ] Dual-framework acceptance criterion (§3.4) passes on at least one pilot college with real mixed-cohort data.
- [ ] RLS penetration test complete, findings closed.
- [ ] DPDP DPA template executed with all pilot colleges; minor-consent flow live and tested with a real under-18 test account.
- [ ] Regression suite over the gold set green on the exact model/engine/framework versions being shipped, versions pinned and recorded.
- [ ] Rollback plan documented and rehearsed for the attainment engine and the migration tool specifically.
- [ ] Support channel and on-call rotation defined for the GA cohort.
- [ ] SAR export for Criteria 1, 3, 8 validated by the Domain/Compliance Lead against the actual official template, table-by-table.

Section 12

Sign-off

This plan is a working draft for kickoff review. Section 3 (dual-framework requirement) is the one section that should not be reinterpreted informally once signed — any change to it should come back through the steering committee, not get absorbed silently into a sprint.

Program Sponsor	Engineering Lead	Domain / Compliance Lead

Source material: AutoOBE v3 Engineering & GTM Plan, verified against the official NBA SAR (nbaind.org, GAPC v4.0, Jan 2025), real AKTU syllabi and question papers, live LLM API pricing, and current DPDP Act rules. Legacy 12-PO text cross-checked against a published NBA-accredited institution's official PO document.